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Critical Review: Chromosome-length Genome Assembly for the Giant Salmonfly *Pteronarcys californica*.

Introduction:

This critical review addresses a study on genome assembly of the giant Salmonfly insect ¹, known as the *Pteronarcys californica* ². Following a decline in its population which has made it endangered, and it has already become regionally extinct in some areas of the United States, and using modern techniques such as PacBio ³, and Hi-C ⁴, to explore the genomic characteristics of this species and to explain its genomic sequencing in order to preserve its survival. And the effectiveness of these tools in supporting the conservation of biodiversity in the long term ¹.

Summary:

The study aims to conserve the giant salmonfly "*P. californica*" by determining its genome sequence and assembly ¹. A single sample was collected from the Diamond Fork River in Spanish Fork Canyon, handled quickly, and cryopreserved using liquid nitrogen at -80°C for DNA extraction ¹. To obtain high-accuracy data, they used PacBio HiFi for long-read sequencing ⁵, and Hi-C to organize the sequences into 14 pseudo chromosomes ⁴, by shearing the DNA into 18 kbp fragments using a specific device. The assembly was performed using HiFiasm software ⁶, for the primary components, then assembled using Juicer ⁷, and 3D-DNA pipeline ⁸, to achieve chromosome length accuracy. They achieved a genome assembly larger than any other stonefly, reaching 2.4 Gb ^{1,9}, which is due to the accumulation of repetitive elements (TEs) ¹⁰.

Critique:

Due to the difficulty of reaching such results with older technologies, the integration of HiFi and Hi-C techniques was a very smart decision ^{3,4}. It made it easier to achieve high accuracy for a large and complex genome of 2.4 Gbp ¹. Despite the sample being old and frozen, the quality of the results was excellent, with a BUSCO score of 98.8%¹¹⁻¹³. This proves that the DNA was handled with care and preserved in good condition. The integration of HiFi and Hi-C technologies improved the assembly

quality^{3,4}; however, the high cost of these methods may limit their widespread use in conservation studies. Also, the study relied on a single sample¹, which may not represent the full genetic diversity of the species. It was not generally recorded how genomic data would actually be used under protection.

Conclusions:

This study has provided an important genomic foundation for bioinformatics in the conservation of endangered species by generating a high-quality genome for an endangered species. However, this study faces challenges, including a high percentage of unclassified repetitive elements REs (approximately 60.6%), accounting for 37.2% of the overall assembly¹. limited practical applications, and high equipment costs. All of these challenges hinder the impact of these results.

Future studies should include multiple samples and integrate genomic data with ecological and conservation strategies to improve the practical impact of the findings.

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